Multi-Agent Receding-Horizon Games for Autonomous Market Participation

Detailed Case-Study Report — Distributed Day-Ahead ADMM & Explicit Real-Time FACET Generalized Nash Equilibrium

Green-H_2 coalition, ERCOT 2025, $H = 4$ (1-hour) receding horizon, PCC band $[100, 900]$ kW

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Contents

Abstract

Six independently-owned green-hydrogen electrolyzers behind one point-of-common-coupling (PCC) bid collectively into the ERCOT two-settlement market. **Both** market stages are solved fully distributed, over the *same* physical PCC import band $[L_{\min}, L_{\max}] = [100, 900]$ kW, and *no agent ever shares its cost model*. The day-ahead (DAM) anchor is set by a distributed **ADMM** in which each agent solves only its own 24-hour QP and the sole shared quantity is the per-hour coalition import. Real-time dispatch (RTM, 15-min) is a **receding-horizon** ($H = 4$) parametric generalized Nash equilibrium (GNE) solved online by **FACET**: each agent locates its own precomputed critical region by warm-started point-location over its facet-neighbour graph (never enumerating the $K^N \approx 1.1 \times 10^{21}$ combinations), the coalition combination is made self-consistent, and the *variational* GNE (= social optimum) is recovered by a potential-minimizing solve that remains valid when the coupling binds. All six agents are distinct in converter rating and efficiency, so each carries its own offline map. Over two independent ERCOT weeks (1–7 April, 7–13 July 2025; −\$1 to \$3553/MWh) FACET clears **99.3%** of steps *iteration-free* — a **median of one inter-agent round per step** against the 67 a tuned distributed ADMM needs for the same equilibrium (1.8% of its communication over the two weeks), at a median **3.9 ms** against a 900 s dispatch interval — meeting the weekly H_2 target at **140–141%**, holding renewable curtailment to **5.8–6.8%** of available generation, and matching the centralized equilibrium to $\sim 10^{-4}$ kW in the interior. Against ADMM head-to-head, identically initialised and judged by the same metric, FACET is $\sim 34\times$ faster at the median step; the robust, implementation-independent claim, though, is the communication (1 vs 67 rounds). On the 0.7% of steps at the degenerate coupling ceiling FACET returns a certified but not-necessarily-*variational* GNE (bounded to $\sim 2\%$ of the PCC band), and a warm-started distributed ADMM fallback (~ 27 rounds) is the only iteration. This report documents each agent's optimization problem, the distributed DAM and the online FACET clearing step by step, and all measured results including the solver comparison, solve times and curtailment. §9 reports honestly that the residual degeneracy — not the fleet composition — is what bounds the iteration-free rate.

1 Nomenclature

Every symbol and every term of art used in this report is defined here, and used nowhere in a looser sense.

Terms of art

Generalized Nash equilibrium (GNE)
A collective decision at which no agent can lower its own cost by unilaterally changing its own decision. "Generalized" means the agents' *feasible sets* are coupled — here by the PCC band — not merely their costs.

Variational GNE (v-GNE)
The particular GNE at which the Lagrange multipliers of the coupling constraint are equal across agents. For a potential game it coincides with the minimizer of the potential, i.e. the *social optimum*. When several GNE exist, this is the one we select.

Potential game
A game admitting a single function Φ (the *potential*) whose gradient reproduces every agent's cost gradient. Consequence used throughout: the v-GNE is $\arg\min \Phi$ over the joint feasible set.

Multiparametric QP (mpQP)
A quadratic program solved *once, offline*, symbolically in its parameters, yielding the optimizer as an explicit piecewise-affine function of those parameters.

Critical region (CR)
One piece of that piecewise-affine function: a polytope in parameter space inside which the optimizer is a single affine function of the parameters. Written $E_i t_i \leq f_i$ with law $x_i = G_i t_i + g_i$.

Combination
A choice of one CR per agent, $\mathcal{C} = (r_0, \ldots, r_5)$. There are $K^N \approx 1.1 \times 10^{21}$ of them; we never enumerate them.

Point location
Determining which CR contains a given parameter point, by the containment test $E_i t_i \leq f_i$. Pure arithmetic: no optimization, no randomness, no tolerance beyond ε.

Warm start
The initial iterate $x^{(0)}$ of a step, taken to be the *previous* 15-minute step's certified solution. Not an approximation of the answer — only a starting point whose correctness is tested (see *self-consistency*).

Iterate
The value $x^{(r)}$ carried at round r of the fixed-point loop. Each round produces the next iterate by an exact solve; iterates are never averaged, relaxed, or damped.

Self-consistency
The acceptance test: recomputing every agent's parameter point from the *solved* decision returns the same CRs that were assumed in order to compute it. Passing this test certifies a GNE (§7.1.6); nothing is accepted without it.

Facet neighbour
A CR sharing a facet (a full-dimensional face) with a given CR.

Iteration-free step
A step cleared with **exactly one decision broadcast per agent**. All location, solving and refinement is local computation on the onboard maps and costs no communication, however many rounds it takes.

Fallback step
A step where the loop fails to certify and a warm-started distributed ADMM is run instead — the only part of the method in which agents exchange messages iteratively.

Symbol	Meaning	Size / value
Indices and dimensions		
i, j	agent indices	$0, \ldots, 5$
N	number of agents	6
k	step index within the horizon	$0, \ldots, H-1$
H	receding horizon length	4 steps (1 hour)
Δt	step duration	0.25 h
t	real-time (RTM) step index within a day	$0, \ldots, 95$
r	round index of the fixed-point loop	$1, \ldots, 4$
Decisions		
p_i	agent i's grid import over the horizon	$\mathbb{R}^4$ (kW)
cv_i	agent i's renewable curtailment (renewable agents only)	$\mathbb{R}^4$ (kW)
x_i	agent i's decision block: p_i (grid) or $[p_i; cv_i]$ (renewable)	$\mathbb{R}^4$ or $\mathbb{R}^8$
x	stacked decision vector of all agents	$\mathbb{R}^{40}$
$x^\star$	the certified equilibrium of a step	$\mathbb{R}^{40}$
Parameters		
θ	**public** market data of the current step	$\mathbb{R}^{24}$
D_i	H_2 still owed by agent i over the horizon	kg
λ_{RT}	real-time price over the horizon	\$/MWh
$p_{\mathrm{DA},i}$	agent i's day-ahead award for the current hour	kW
g_s, g_w	solar / wind availability at the shared farms	kW
s_i	coalition aggregate excluding i: $\sum_{j \neq i} p_j$	$\mathbb{R}^4$ (kW)
t_i	agent i's **private** parameter point $[\, s_i \,;\, \theta \,]$	$\mathbb{R}^{28}$
S	selector: extracts the first H entries (the p-part) of x_j	$\mathbb{R}^{4 \times n_{x_j}}$
Physical and economic constants		
$L_{\min}, L_{\max}$	PCC import band, binding on $\sum_i p_i$ at every step	$100,\ 900$ kW
$P_{\max,i}$	agent i's electrolyzer (converter) rating	kW
η_i	agent i's electrolyzer efficiency	kg/kWh
Maps and equilibrium system		
E_i, f_i	the polytope of one CR of agent i	$E_i t_i \leq f_i$
G_i, g_i	the affine law carried by that CR	$x_i = G_i t_i + g_i$
G_i^s, G_i^θ	G_i split at column H, matching $t_i = [s_i; \theta]$	—
r_i	index of the CR located for agent i	—
$\mathcal{C}$	combination: one CR per agent, $(r_0, \ldots, r_5)$	—
M_x, M_p, M_1	the equilibrium system $M_x x = M_p \theta + M_1$	$40{\times}40$, $40{\times}24$, $\mathbb{R}^{40}$
V_2	basis of $\ker M_x$ when M_x is rank-deficient	—
y_2	coordinate on the GNE manifold, $x = x_p + V_2 y_2$	$\mathbb{R}^{40-\operatorname{rank} M_x}$
Cost		
Q, c, F	stacked cost: $\Phi(x) = \frac{1}{2} x^\top Q x + (c + F\theta)^\top x$	$Q \succ 0$, $\lambda_{\min} = 10^{-3}$
Φ	coalition potential; its minimizer over the feasible set is the v-GNE	—
G, w_0, W	joint feasible set of the coalition, $Gx \leq w_0 + W\theta$	142×40

Table 1: Nomenclature. Sizes are those of the case study of this report (N=6, H=4).

2 Method overview and contributions

This section states the problem, the method, and what is new, in full but without detail. The complete derivation, line-by-line algorithm and two traced numerical examples are in §7.1; readers who want the mechanism should go there.

2.1 The problem

Six independently-owned green-hydrogen electrolyzers sit behind one point of common coupling (PCC) and bid as a coalition into ERCOT. Every 15 minutes the market publishes θ — prices, renewable availability, each agent's remaining H_2 obligation, the day-ahead awards — and the coalition must produce a dispatch $x^\star$ that is a **generalized Nash equilibrium**: no agent can lower its own cost by deviating, and the coalition's total import obeys the physical band $L_{\min} \le \sum_i p_i \le L_{\max}$ at every step.

The binding difficulty is not the equilibrium itself but the ownership model. **No agent discloses its cost model**, so no party — not even a hypothetical coordinator — can assemble the joint problem at runtime. Any admissible method must be built from quantities each agent can compute alone, plus data that is already public. The only such coupling quantity is the coalition's total import, which is already measured at the PCC meter.

2.2 What FACET-RHG does

Offline and once, each agent solves *its own* mpQP and stores the result as a lookup table of critical regions. Online, at each step, the coalition runs a **fixed-point iteration on the combination**, summarized in Figure 1.

One 15-minute step: warm start → locate → solve → certify

0. **Warm start.** Take $x^{(0)}$ to be the previous step's *certified* solution. Steps are 15 minutes apart, so θ has moved little.
1. **Locate** (each agent, locally, in parallel). Agent i forms its private parameter point $t_i = [\,s_i\,;\,\theta\,]$, where $s_i = \sum_{j\neq i} p_j$ is read off the current iterate, and finds the CR containing it by the containment test $E_i t_i \le f_i$. Deterministic arithmetic — no optimization, no randomness. Six lookups give a combination $\mathcal{C}$.
2. **Solve exactly.** *Given* $\mathcal{C}$, every agent's law is affine, so the six coupled best responses collapse into one linear system $M_x x = M_p\theta + M_1$ whose exact solution returns *all six* agents' decisions at once. Where M_x is rank-deficient — the coupling ceiling — the GNE form a manifold and we select the *variational* one by minimizing the potential Φ over it.
3. **Certify.** Recompute each t_i from the *solved* x and locate again. If every agent returns the CR it was assumed to be in, the combination has reproduced itself: each agent is provably best-responding to the aggregate that actually materializes, which *is* the definition of a GNE. Accept and stop. Otherwise adopt x as the next iterate and return to 1.

Steps 1–3 are **local computation on the onboard maps**. The inter-agent exchange is **exactly one decision broadcast per agent per step**, regardless of how many rounds run.

Figure 1: The online method. Nothing is accepted that has not passed step 3.

Two properties are worth stating plainly, because both are easy to mis-read:

- **The warm start is not an approximation of the answer.** It is only a starting point, and its correctness is *tested* in step 3. A poor warm start costs extra rounds of local arithmetic, never a wrong dispatch. On 6 April, 82 of 96 steps certified on the first round.
- **The method is sound, not complete.** Step 3 never accepts a wrong answer, but the loop is not guaranteed to reach a fixed point. When it does not, the step falls back (§7); the cost is a slower step, never a silently incorrect one.

2.3 What is new here

The architecture — per-agent offline maps, a block linear system of affine best responses, membership validation, one exchange per step — is inherited from our IF-mpDiMPC work (Saini *et al.*, *Comput. Chem. Eng.* 200, 2025) and its ACC-2026 successor. **Point location is not new: IF-mpDiMPC-V2 already performs it.** What is new is what the *game* forces, none of which arises in cooperative distributed MPC, where all controllers share one objective and the solution is unique:

1. **Rank-deficient systems carry the answer instead of signalling an error.** IF-mpDiMPC discards singular systems (`if abs(det(L)) < 1e-10:  return None`) — correct there, because a unique solution means singularity implies a wrong combination. Here M_x is genuinely rank-deficient whenever the PCC band binds for several agents at once (rank 36 of 40 in the traced example of §7.1.8), the equilibria form a manifold, and the answer lies on it. We select the variational GNE from that manifold, and re-impose feasibility on the selection when required (Tier 1c).
2. **The iterate is refined rather than enumerated around.** IF-mpDiMPC-V2 and FACET-DiMPC point-locate once and compensate by enumerating the product $\prod_i(1 + |\mathcal{N}_i|)$ over each agent's facet neighbours — 985,608 combinations at t=0 for this fleet. We instead re-locate and iterate, solving a mean of **1.2** combinations per step. This matters because the locality assumption underpinning the neighbour product fails in this game: when the combination changes, 93.1% (April) / 92.1% (July) of the changes are more than three facet-hops away (§7.1.10).
3. **The coupling is sum-compressed.** IF-mpDiMPC's parameter vector carries every other controller's full input trajectory, growing as $M \cdot N_p \cdot n_u$; ours carries four numbers, because the PCC band is the only coupling. This is simultaneously the scalability argument and the privacy argument — the coalition total is already on the meter.

2.4 Headline results

Over two independent ERCOT weeks (1–7 April and 7–13 July 2025; prices −\$1 to \$3553/MWh; 1344 real-time steps):

Iteration-free steps	**99.3**% (9 fallbacks of 1344)
Communication on a map step	exactly 1 decision broadcast per agent
Median real-time step	**125** ms against a 900 s dispatch interval
Agreement with the centralized equilibrium	$\sim 10^{-4}$ kW in the interior; 5.9 kW worst case
Combinations ever visited	700 of $K^N \approx 1.1 \times 10^{21}$ (6.5×10^{-19})
Weekly H_2 target	met at 140–141%
Renewable curtailment	5.8–6.8% of available generation

§9 reports honestly that the residual degeneracy at the coupling ceiling — not the fleet composition — is what bounds the iteration-free rate, and §10 states which comparisons in this report are not yet on an equal footing.

3 Physical system and market

A cluster of small electrolyzers sits behind one shared PCC. Each owner runs its own asset for its own revenue and keeps its cost model private; only power decisions are shared. Two forces make this a game rather than N independent problems:

- **Physical coupling.** The PCC caps the collective import and a market floor sets a minimum offer, *at every 15-min step k*:

$$L_{\min} = 100 \text{ kW} \;\leq\; \sum_{i=1}^{N} p_{i,k} \;\leq\; L_{\max} = 900 \text{ kW}.$$

  $L_{\min}$ is the ISO minimum offer size; $L_{\max}$ is the physical interconnection limit (75% of the 1200 kW fleet nameplate). **This same band governs both the day-ahead and the real-time stage** — there is one physical wire.
- **Private hydrogen contracts.** Each agent must produce a daily H_2 quantity; it may time production freely but the daily total is a hard floor. That freedom, under a shared per-step power limit, is what couples the agents and the horizon.

The market is ERCOT two-settlement: a day-ahead (DAM, hourly) stage that sets a financial anchor p^{DA}, and a real-time (RTM, 15-min SCED) stage that re-optimizes against live prices. Electricity $\to$ H_2 conversion: kg $= \eta_i \, p_{\mathrm{elec},i} \, \Delta t$, $\Delta t = 0.25$ h.

4 The fleet (medium-scale)

i	Agent	type	$P_{\max,i}$ [kW]	farm cap [kW]	η_i [kg/kWh]	γ_i	break-even a_i [$/MWh]
0	PEM_Elec	grid	250	–	0.0200	5×10^{-3}	60.0
1	ALK	grid	200	–	0.0180	5×10^{-3}	54.0
2	PEM_PV	PV	150	125	0.0210	5×10^{-3}	63.0
3	PEM_PV_2	PV	100	125	0.0190	5×10^{-3}	57.0
4	PEM_Wind	wind	275	250	0.0195	4×10^{-3}	58.5
5	PEM_Wind_2	wind	225	250	0.0185	4×10^{-3}	55.5

Table 2: $N = 6$, total electrolyzer nameplate $\sum_i P_{\max,i} = 1200$ kW against $L_{\max} = 900$ kW, so the coupling ceiling binds. $r_{\mathrm{H2}} = \$3$/kg, $H = 4$, $\Delta t = 0.25$ h. $P_{\max,i}$ is the *electrolyzer* rating; "farm cap" is the co-located renewable array, which is shared within a technology (both PV agents see the same irradiance $g_h = \mathrm{CF}_h \cdot 125$ kW, both wind agents the same $\mathrm{CF}_h \cdot 250$ kW). Break-even $a_i = r_{\mathrm{H2}}\eta_i \cdot 10^3$: below a_i buying grid power to make H_2 is profitable, above it the agent wants to stop. **All six agents are distinct in** $(P_{\max,i}, \eta_i)$, so all six a_i are distinct and each agent requires its own offline solve. Agents 3 and 5 are inverter-limited (farm $> P_{\max}$) and therefore curtail 25 kW at peak resource regardless of price.

Distinct efficiencies remove one degeneracy but do not improve the iteration-free rate

An earlier version of this study used two *identical* PV agents and two identical wind agents. Identical agents are ill-posed at the coupling ceiling in an obvious way: if agents j and k have the same cost and constraints, any split of their combined power is equally optimal, so a continuum of GNE exists. Making every η_i distinct strictly orders the marginal H_2 values $a_i = r_{\mathrm{H2}}\eta_i$, so "who backs off first" has a unique answer and *that* tie disappears by construction.

Measured, this alone did not help. Holding the clearing algorithm fixed, the identical-pair fleet was 99.2% iteration-free (11/1344 ADMM fallbacks) while the distinct fleet was 97.2% (37/1344). Permutation symmetry was never the *binding* source of degeneracy: per Hall & Bemporad (Def. 2), M_x loses rank whenever the *shared coupling constraint* is active for two or more agents at once — which happens regardless of whether those agents are identical. Removing the symmetry eliminated a secondary tie while the dominant one remained, and the distinct fleet additionally brought richer maps and a new degenerate surface (forced curtailment).

Fixing the real cause recovers it, in two steps. The dominant degeneracy is handled by the feasibility-constrained variational selection of §7 (Tier 1c), which addresses the shared constraint rather than the fleet: with it alone the distinct fleet reached 98.8% (16/1344). A second, independent fix then closed the rest of the gap: the offline facet-neighbour graph feeding Tiers 1b/1c was itself badly incomplete (§7, "`facet_crossing`"); rebuilding it correctly brought the distinct fleet to 99.3% (9/1344) — matching the identical-pair fleet's 99.2%. *Fleet composition was never the lever; equilibrium selection and a complete neighbour graph are.*

5 Day-Ahead stage: distributed ADMM (model-private)

The day-ahead schedule is solved once per day on the DAM *forecast* ($\hat{\lambda}_h = \lambda_h^{\mathrm{DA}} + \mathcal{N}(0, 16)$, renewable $\hat{g}_{i,h}$). It is **distributed**: each agent i solves only its own 24-hour problem, and the *only* exchanged quantity is the per-hour coalition import $\sum_i p_{i,h}$ (already public at the metered point). No agent reveals $P_{\max,i}$, η_i, its renewable cap, or its daily target.

Per-agent day-ahead problem. With $\Delta t_{\mathrm{DA}} = 1$ h and daily target $D_i^{\mathrm{day}} = 0.55\, P_{\max,i}\eta_i \cdot 24$, agent i solves

$$\min_{p_{i,h}\geq 0,\; cv_{i,h}\geq 0} \sum_{h=0}^{23} \Big[(\hat{\lambda}_h/1000 - r_{\mathrm{H2}}\eta_i)\,\Delta t_{\mathrm{DA}}\, p_{i,h} + r_{\mathrm{H2}}\eta_i\,\Delta t_{\mathrm{DA}}\, cv_{i,h} + \underbrace{\tfrac{1}{2}\gamma_{\mathrm{DA}}\, p_{i,h}^2}_{\text{regularizer}} \Big]$$

$$\begin{aligned}
&0 \leq p_{i,h} \leq P_{\max,i}, \quad 0 \leq cv_{i,h} \leq \mathrm{cap}_i && \text{(local box; } cv \text{ only renewables)}\\
&0 \;\leq\; p_{i,h} + \hat{g}_{i,h} - cv_{i,h} \;\leq\; P_{\max,i} && \text{(local renewable balance)}\\
&\textstyle\sum_{h=0}^{23} \eta_i\,(p_{i,h} + \hat{g}_{i,h} - cv_{i,h})\,\Delta t_{\mathrm{DA}} \;\geq\; D_i^{\mathrm{day}} && \text{(hard daily } \mathrm{H_2} \text{ floor)}
\end{aligned}$$

Every term uses only agent i's own data; the daily H_2 floor couples that agent's own 24 hours. The **coupling band** $L_{\min} \leq \sum_i p_{i,h} \leq L_{\max}$ $(\forall h)$ is the *only* cross-agent constraint, enforced by ADMM consensus/dual updates that see nothing but the per-hour aggregate:

$$\text{(x) } p_i^{\ell+1} = \arg\min\big[J_i(p_i) + \tfrac{\rho}{2}\|p_i - z_i^\ell\|^2\big], \quad \text{(z) } z^{\ell+1} = \Pi_{[L_{\min}, L_{\max}]}\big(\textstyle\sum_i p_i^{\ell+1} + \mu^\ell/\rho\big), \quad (\mu)\ \mu_i^{\ell+1} = \mu_i^\ell + \rho(p_i^{\ell+1} - z_i^{\ell+1}),$$

with $\rho = 0.1$; Π is a closed-form clip of the per-hour aggregate onto the band. The award $p_{i,h}^{\mathrm{DA}}$ is the RTM anchor; its cumulative-H_2 trajectory sets the RTM pacing target $D_i(t)$. (`dam.solve_dam_admm`; consensus in `admm_solver.py`.)

The small γ_{DA} regularizer (why, and its trade-off)

The day-ahead economics are *linear* in each hourly buy: "buy enough cheap power to make the day's hydrogen." With no γ_{DA} the problem is a degenerate LP — on a cheap day it says *buy the maximum every hour*, a knife-edge answer that ADMM oscillates on and cannot settle. A *tiny* quadratic penalty $\frac{1}{2}\gamma_{\mathrm{DA}} p_{i,h}^2$ ($\gamma_{\mathrm{DA}} = 10^{-4}$) rounds the knife-edge: it spreads purchases smoothly, gives a *unique* schedule so ADMM converges, and changes total cost by $\leq$1.4%. A matching $\varepsilon = 10^{-3}$ regularizes renewable curtailment cv. Validation: a centralized solve of the *same* regularized problem (offline oracle, `solve_dam_centralized`) is matched to $\leq$10 kW on priced days; on ultra-flat days the day-ahead problem is genuinely degenerate so the two solvers land on different equally-optimal schedules ($\leq$23 kW apart, cost within 1%).

6 Real-Time stage: each agent's mpQP

Every RTM step, agent i solves a strictly-convex mpQP over the horizon $k = 0, \ldots, 3$. The decision is grid import $p_{i,k} \geq 0$ (renewable agents additionally choose curtailment $cv_{i,k}$). The cost is a day-ahead deviation anchor $\frac{1}{2}\gamma_i(p - p_i^{\mathrm{DA}})^2$ (strict convexity, $Q_i \succ 0$) plus real-time energy cost net of H_2 revenue. **Each agent's cost and every local constraint use only that agent's own parameters; the sole shared quantity is the aggregate $\mathrm{s}_k = \sum_{j \neq i} p_{j,k}$.**

6.1 Grid agents (PEM_Elec i=0, ALK i=1)

Grid agent i — decision $x_i = [p_{i,0..3}] \in \mathbb{R}^4$

Private parameters $\theta_i = [\,\mathrm{s}_0..\mathrm{s}_3 \mid D_i,\ \lambda_0..\lambda_3,\ p_i^{\mathrm{DA}}\,] \in \mathbb{R}^{10}$

$$\min_{p_i \in \mathbb{R}^4} \sum_{k=0}^{3} \Big[\tfrac{1}{2}\gamma_i\,(p_{i,k} - p_i^{\mathrm{DA}})^2 + \big(\lambda_k/1000 - r_{\mathrm{H2}}\eta_i\big)\,\Delta t\; p_{i,k}\Big]$$

$$\begin{aligned}
&0 \leq p_{i,k} \leq P_{\max,i}, && \text{(capacity box)}\\
&\textstyle\sum_k \eta_i\, p_{i,k}\,\Delta t \ \geq\ D_i, && \text{(cumulative } H_2 \text{ — couples the 4 steps)}\\
&L_{\min} \ \leq\ p_{i,k} + \mathrm{s}_k \ \leq\ L_{\max}. && \text{(shared per-step coupling)}
\end{aligned}$$

6.2 Renewable agents (PEM_PV i=2, 3; PEM_Wind i=4, 5)

Renewable output $g_{i,k}$ feeds the electrolyzer load $p_{\mathrm{elec},i,k} = p_{i,k} + g_{i,k} - cv_{i,k}$ (free renewable displaces grid import; curtailment cv throws away H_2 at break-even).

Renewable agent i — decision $x_i = [p_{i,0..3},\, cv_{i,0..3}] \in \mathbb{R}^8$

Private parameters $\theta_i = [\,\mathrm{s}_0..\mathrm{s}_3 \mid D_i,\ \lambda_0..\lambda_3,\ p_i^{\mathrm{DA}},\ g_0..g_3\,] \in \mathbb{R}^{14}$

$$\min_{p_i, cv_i} \sum_{k=0}^{3} \Big[\tfrac{1}{2}\gamma_i(p_{i,k} - p_i^{\mathrm{DA}})^2 + \tfrac{1}{2}\varepsilon\, cv_{i,k}^2 + (\lambda_k/1000 - r_{\mathrm{H2}}\eta_i)\Delta t\, p_{i,k} + r_{\mathrm{H2}}\eta_i\,\Delta t\, cv_{i,k}\Big]$$

$$
\begin{aligned}
&0 \le p_{i,k} \le P_{\max,i}, \quad 0 \le cv_{i,k} \le g_{i,k} \le \mathrm{cap}_i,\\
&0 \;\le\; p_{\mathrm{elec},i,k} = p_{i,k} + g_{i,k} - cv_{i,k} \;\le\; P_{\max,i},\\
&\textstyle\sum_k \eta_i\, p_{\mathrm{elec},i,k}\,\Delta t \;\ge\; D_i, \qquad L_{\min} \;\le\; p_{i,k} + \mathrm{s}_k \;\le\; L_{\max} \text{ (grid import only).}
\end{aligned}
$$

Receding-horizon coupling (not a repeated static game). The four steps couple only through the cumulative-H_2 row. At an H_2-binding point (PEM, raise λ_1 by \$40/MWh): $\Delta p = [+0.5, -1.5, +0.5, +0.5]$ kW with $\sum_k p_k : 600 \to 600$ (H_2-conserved). Step 1 becomes expensive so it drops; the others rise to keep the H_2 total — a single-step map would give $\Delta p = \mathbf{0}$.

Sum-compression and elimination. Agent i depends on the others only through the per-step aggregate $\mathrm{s}_k = \sum_{j \ne i} p_{j,k}$ (4 numbers), not their private $D_j, p_j^{\mathrm{DA}}, g_j$. Each agent's mpQP is solved offline (PPOPT) in its private 10- or 14-dimensional θ_i; the coupling variable is eliminated when the coalition combination is assembled. Public parameter count is $4(\lambda)+6(p^{\mathrm{DA}})+6(D)+8(g) = 24$, but no 24-dimensional problem is ever solved.

The daily pacing target is receding: $D_i(t) = \mathrm{clip}\big((D_i^{\mathrm{day}} - h2_i^{\mathrm{inv}})\, H/\text{steps left in day}, 0, D_i^{\max}\big)$; only step $k = 0$ is executed against realized price and renewable, then the window advances.

7 Online FACET clearing of the RTM GNE

Each agent's offline solution is an explicit piecewise-affine map: a set of *critical regions* (CRs), each a polytope $E_i\theta_i \le f_i$ carrying an affine law $x_i = G_i\theta_i + g_i$. A *combination* $\mathcal{C} = (r_0, \ldots, r_5)$ picks one CR per agent; the equilibrium at combination $\mathcal{C}$ solves a block-linear system $M_x x = M_p\theta + M_1$. FACET finds and solves the correct combination online **without enumerating the $K^N \approx 1.1\times10^{21}$ combinations**.

7.1 How FACET works, step by step

This subsection builds the online method from nothing, in the order the code executes it, and then runs two *real* 15-minute steps end to end with the actual numbers. Every figure quoted is a live trace of the shipped code on 6 April 2025 — none of it is illustrative.

1. The question we must answer every 15 minutes

At each real-time step the market hands us θ — 24 public numbers: the four upcoming RT prices, the six agents' H_2 still owed, the six DAM awards, and the solar/wind available to each farm. From θ alone we must produce

$$x^\star = \big(x_0, \ldots, x_5\big), \qquad x_i = \text{agent } i\text{'s power over the next 4 steps,}$$

such that **(a)** no agent can reduce its own cost by unilaterally changing its own x_i, and **(b)** the coalition respects the physical PCC band $L_{\min} \le \sum_i p_i \le L_{\max}$ at every step. That is a *generalized Nash equilibrium*. The word "generalized" is doing real work: condition (b) couples the agents' *feasible sets*, not merely their costs.

The hard constraint on any method is §7's privacy model: **no agent reveals its cost model**, so no party — not even a hypothetical coordinator — is able to write down the joint problem at runtime. Whatever we do must be assembled out of things each agent can compute alone.

2. What each agent stores offline: a lookup table

Before the day begins, each agent i solved *its own* mpQP once, and stored the answer as a lookup table. The table has one row per **critical region** (CR). Each row holds

a polytope	$E_i\, t_i \le f_i$	— "does this row apply?"
an affine law	$x_i = G_i\, t_i + g_i$	— "if it applies, here is my exact optimum"

Using the table is not an optimization. It is a lookup with two operations: find the row whose polytope contains the query point t_i (a containment test $E_i t_i \le f_i$ — pure arithmetic, no solver, no randomness), then evaluate one affine formula. The tables are big but they are only tables: 507 rows for PEM_Elec up to 12,782 for PEM_PV_2, 35,348 in total (Table 3).

We call the choice of one row per agent a **combination** $\mathcal{C} = (r_0, \ldots, r_5)$. If we knew the right combination, we would be finished — that is the entire point of having the tables.

3. The one subtle part: what the query point contains

What is t_i — the key we look the table up with? It is *not* just θ:

$$t_i \;=\; \Big[\; \underbrace{\textstyle\sum_{j\neq i} p_j}_{\text{4 numbers}} \;\Big|\; \underbrace{\theta}_{\text{24 numbers}} \;\Big] \in \mathbb{R}^{28}.$$

Why does agent i's own optimum depend on the others at all? **Only through the PCC band.** Agent i may draw whatever it likes provided $\sum_j p_j$ stays inside $[L_{\min}, L_{\max}]$ — so the others constrain i *exclusively via their total*, never individually. That is why four numbers suffice (one per horizon step), and it is also the whole privacy argument: the coalition total is already visible at the PCC meter, so sharing it reveals nothing. Agent i never sees any $P_{\max,j}$, η_j, or cost.

This is also precisely what makes the problem a *game* rather than six independent problems.

4. The circularity — the only genuinely hard part

The difficulty in one sentence

To look up my row I need $\sum_{j\neq i} p_j$. But $\sum_{j\neq i} p_j$ is part of the answer I am trying to compute. **The key depends on the value.**

Everything else in FACET is machinery for breaking this circle without a central solver. If this subsection is confusing, it is almost certainly *this* point that is the source — the rest is bookkeeping around it.

5. Breaking the circle: warm start → locate → solve → certify

The circle is broken by a **fixed-point iteration that carries its own proof of correctness**.

Algorithm 1 One FACET clearing step (Tier 1). In: θ, previous solution x^{prev}, previous combination $\mathcal{C}^{\text{prev}}$. Out: a certified GNE $x^\star$.

1: $x \leftarrow x^{\text{prev}};\quad \mathcal{C} \leftarrow \mathcal{C}^{\text{prev}}$ ▷ **warm start**: previous step's certified solution
2: **for** round $= 1, \dots, 4$ **do**
3:     **for** each agent $i = 0, \dots, 5$ **do** ▷ local & parallel; no messages
4:         $t_i \leftarrow \big[\, \sum_{j \neq i} p_j(x) \;\big|\; \theta \,\big]$ ▷ form my key from the current iterate
5:         $r_i \leftarrow \textsc{Locate}\big(t_i,\ \text{hint} = \mathcal{C}_i\big)$ ▷ **locate** — which row holds t_i?
6:     **end for**
7:     $\mathcal{C}' \leftarrow (r_0, \dots, r_5)$
8:     $x' \leftarrow \textsc{SolveExact}(\mathcal{C}', \theta)$ ▷ **solve** — $M_x x = M_p \theta + M_1$, all six at once
9:     **if** $\textsc{Locate}\big([\, \sum_{j \neq i} p_j(x') \,|\, \theta \,]\big) = r_i$ for every i **then**
10:         **return** x' ▷ **re-check passed** $\Rightarrow$ certified GNE; stop
11:     **end if**
12:     $x \leftarrow x';\quad \mathcal{C} \leftarrow \mathcal{C}'$ ▷ someone crossed a boundary — iterate
13: **end for**

Before any example, here is **every line of Algorithm 1, with the exact operation it performs**. Throughout, $x \in \mathbb{R}^{40}$ is the stacked decision vector and $\theta \in \mathbb{R}^{24}$ the public data:

i	agent	n_{x_i}	block x_i and its slot in x	
0	PEM_Elec	4	$p_i \in \mathbb{R}^4$	$x[0{:}4]$
1	ALK	4	$p_i \in \mathbb{R}^4$	$x[4{:}8]$
$2, 3$	PEM_PV, PEM_PV_2	8	$[\, p_i\, ;\, cv_i \,] \in \mathbb{R}^8$	$x[8{:}16]$, $x[16{:}24]$
$4, 5$	PEM_Wind, PEM_Wind_2	8	$[\, p_i\, ;\, cv_i \,] \in \mathbb{R}^8$	$x[24{:}32]$, $x[32{:}40]$
$n_x^{\text{tot}} = 2 \cdot 4 + 4 \cdot 8 = \mathbf{40}$,		$n_\theta = \mathbf{24}$,	$H = 4$	

Line 1 — the warm start ($x \leftarrow x^{\text{prev}}$, $\mathcal{C} \leftarrow \mathcal{C}^{\text{prev}}$). We begin from the previous 15-minute step's certified answer and the rows it used. *At the very first step of a day there is no previous step*: the code seeds x with the DAM award, $p_i(k) \leftarrow p_{\text{DA},i}$ for every k, and sets $\mathcal{C} \leftarrow \varnothing$ — so line 5 has no hint and performs a full scan exactly once per day. Every later step inherits the previous one.

Line 2 — the round loop. At most 4 rounds. One point is essential and easy to miss: θ **is fixed for the whole loop.** It is the data of *this* step and does not change between rounds. The *only* thing that changes from round to round is the iterate x, and therefore the aggregates s_i built from it.

Line 3 — the per-agent loop. Every agent executes lines 4–5 on its own hardware, in parallel, using only its own table. No messages are exchanged inside the loop.

Line 4 — form the lookup key, $t_i \leftarrow [\, \sum_{j \neq i} p_j(x) \mid \theta \,]$. Two concatenated pieces. First the aggregate of everyone else, read off the *current iterate* x:

$$s_i \;=\; \sum_{j \neq i} p_j \;=\; \sum_{j \neq i} S\, x_j \;\in\; \mathbb{R}^4, \qquad S x_j := \text{first } H{=}4 \text{ entries of } x_j \text{ (the grid-import part)},$$

(for renewable agents S discards the curtailment block cv_j — only imported grid power crosses the PCC). Then θ is appended:

$$t_i = \big[\, s_i \,;\, \theta \,\big] \in \mathbb{R}^{4+24} = \mathbb{R}^{28}.$$

Which θ is this? *This step's θ* — rebuilt from scratch at every RTM step out of the current market data: the new RT prices λ_{RT}, the freshly updated H_2 still owed D_i, the current solar/wind availability g, and (on the hour) the new DAM award p_{DA}. It is *public* and therefore identical for all six agents; only the first four entries of t_i differ between agents. And per line 2, it is constant across rounds.

Line 5 — locate, $r_i \leftarrow \text{Locate}(t_i,\ \text{hint} = \mathcal{C}_i)$. The question asked is only: *which row r of my table satisfies $E_r\, t_i \le f_r$?* The procedure is

(a) test the hint row: if $\max(E_{\mathcal{C}_i} t_i - f_{\mathcal{C}_i}) \le \varepsilon$, return $\mathcal{C}_i$;
(b) else test the hint's facet neighbours, return the first that contains t_i;
(c) else scan all rows with one batched matrix product, viol $= \max_{\text{rows}}\big(\mathbf{E}\, t_i - \mathbf{f}\big)$, and return the first index with viol $\le \varepsilon$.

What "hint" means on line 5 — and why it is safe

The **hint** is simply *the row this agent used last time*: on round 1 it is the row from the previous 15-minute step; on rounds $2, 3, 4$ it is the row found in the previous round. It is carried in $\mathcal{C}$.
The hint is **purely a speed device — it cannot change the answer, only the time taken to find it.** Whenever t_i lies strictly inside a region, that region is the unique one containing it, so (a), (b) and (c) all return the same row. If t_i lands exactly on a shared facet, two rows contain it and (a)–(c) may return different indices — but the piecewise-affine map is *continuous*, so both rows' affine laws produce the identical x_i. Either answer is correct.
This is why the warm start can never corrupt the result: a bad hint costs a few wasted containment tests, never a wrong equilibrium. Measured: the hint alone answers 86.1% (April) / 77.3% (July) of lookups — one test against 19–49 inequality rows (§9).

Line 7 — assemble the combination, $\mathcal{C}' \leftarrow (r_0, \ldots, r_5)$. Pure bookkeeping: collect the six located row indices into one tuple.

Line 8 — `SolveExact`: the line where the coupling is resolved. This is the heart of the method, so it is worth deriving rather than asserting. Row r_i of agent i's table states its *exact* optimum as an affine function of its key:

$$x_i \;=\; G_i\, t_i + g_i.$$

Split G_i at column H=4, matching the two pieces of t_i:

$$G_i = \big[\; \underbrace{G_i^s}_{n_{x_i}\times 4} \;\big|\; \underbrace{G_i^\theta}_{n_{x_i}\times 24} \;\big] \qquad \Longrightarrow \qquad x_i \;=\; G_i^s\, s_i \;+\; G_i^\theta\, \theta \;+\; g_i.$$

Now substitute what s_i actually is — a linear function of the *other agents' unknowns*, $s_i = \sum_{j\neq i} S\, x_j$ — and move it to the left:

$$\boxed{x_i \;-\; G_i^s \sum_{j\neq i} S\, x_j \;=\; G_i^\theta\, \theta + g_i} \qquad \text{one block row, for each } i = 0, \ldots, 5. \tag{1}$$

Equation (1) is the essential step: the circularity of §4 has become an ordinary *linear* coupling, because inside a known combination every agent's response is affine. Stacking the six block rows gives

$$M_x\, x \;=\; M_p\, \theta + M_1,$$

$$\underbrace{\big[M_x\big]_{ii} = I_{n_{x_i}}}_{\text{“what I do with my own } x_i\text{”}}, \qquad \underbrace{\big[M_x\big]_{ij} = -\,G_i^s S \;\;(j \neq i)}_{\text{“how } j\text{'s import pulls me”}}, \qquad \big[M_p\big]_i = G_i^\theta, \qquad \big[M_1\big]_i = g_i,$$

with $M_x \in \mathbb{R}^{40\times 40}$, $M_p \in \mathbb{R}^{40\times 24}$, $M_1 \in \mathbb{R}^{40}$. The $-G_i^s S$ block sits in the first four columns of agent j's slot — the only columns i can *see*. Read structurally, $M_x = I - (\text{coupling})$. **Six mutually-dependent best responses have collapsed into one linear solve that returns all six agents' decisions at once** — no iteration, no message, no tolerance.

Generic case (M_x invertible). $x' = M_x^{-1}(M_p\theta + M_1)$: the unique GNE of this combination. Example A is this case (40×40, rank 40).

Degenerate case (M_x singular). Example B is this case (rank 36). It happens when the PCC band binds for several agents simultaneously: their laws all reduce to “I take whatever the band leaves me”, so the block rows become linearly dependent. The solutions then form a $\big(40 - \operatorname{rank} M_x\big)$-dimensional *manifold*, obtained from the SVD of M_x:

$$x = x_p + V_2\, y_2, \qquad V_2 = \text{null-space basis of } M_x, \qquad y_2 \in \mathbb{R}^{40-\operatorname{rank} M_x}.$$

Every point on it is a GNE. We select the **variational** one — the social optimum — by minimizing the coalition potential Φ over the manifold:

$$y_2^\star = \arg\min_{y_2}\; \Phi\big(x_p + V_2 y_2\big), \qquad \Phi(x) = \tfrac{1}{2} x^\top Q x + q^\top x, \quad q = c + F\theta,$$

which is a tiny unconstrained QP of dimension 1–4, solved from its normal equations

$$\big(V_2^\top Q V_2\big)\, y_2 \;=\; -\,V_2^\top\big(Q x_p + q\big).$$

This is why a plain `numpy.linalg.solve` is not enough: on Example B it would simply fail. (When even this selection leaves the feasible set, Tier 1c re-imposes the joint feasibility rows on the same minimization — see the keybox in §7.)

Line 9 — the re-check (the acceptance test). Take the freshly computed x', recompute each agent's aggregate *from it*, $s_i' = \sum_{j\neq i} S\, x_j'$, rebuild $t_i' = [\, s_i' \,;\, \theta \,]$ — note θ is unchanged, only the aggregate is new — and locate once more. Accept **only** if all six agents return the same rows r_i they were assumed to be in. §6 explains why passing this test is a proof.

Line 10 — return. The combination reproduced itself: x' is a certified GNE. Stop.

Line 12 — iterate. At least one agent's key crossed a facet, so the affine laws used to build x' were the wrong ones. Adopt x' as the next iterate and repeat. In practice this converges immediately: on 6 April, 82 of 96 steps passed on round 1, 12 needed 2, one needed 3, one needed 4.

Honesty: the loop is not guaranteed to converge

Nothing above proves the fixed point is reached, and it is not always. That is exactly why the round count is capped at 4 and why Tiers 1b/1c/2 exist. What *is* guaranteed is soundness, not completeness: line 9 never accepts a wrong answer, so a failure to converge costs a fallback, never a silently incorrect dispatch. Over both weeks the loop closes on 99.3% of steps.

6. Why self-consistency certifies a generalized Nash equilibrium

Suppose the loop returns x'. The test that let it return says: *when every agent recomputes its key from the aggregate that x' actually produces, it lands in the very row that was used to compute x'.* Since that row's affine law *is* agent i's exact optimum for that key, each x'_i is agent i's exact best response to the aggregate that actually materializes. That is the definition of a generalized Nash equilibrium — so a self-consistent combination is not "close enough", it **is** the equilibrium.

This also explains why nothing may be accepted without the test. A combination that has not reproduced itself is a combination whose affine laws were evaluated at a key that never occurred; its output is meaningless, however plausible the numbers look. Example B below is exactly this trap.

7. Example A — an ordinary step (t=1, 00:15): one round

Market data: prices $\lambda_{\text{RT}} = [11.84, 11.28, 13.63, 13.58]$ \$/MWh (cheap night), no sun ($g_s = 0$), wind $g_w = 168.7$ kW, DAM awards $p_{\text{DA}} = [250, 200, 150, 100, 75, 25]$ kW. Warm start: last step's $\Sigma p = 805.67$ kW at combination $(369, 325, 4001, 9222, 8, 14)$.

Locate. Each agent forms its own key and tests rows:

i	agent	$\sum_{j\neq i} p_j$ (kW)	row found	# ineq.	$\max(E_i t_i - f_i)$
0	PEM_Elec	555.67	369 / 507	19	-2.6×10^{-8}
1	ALK	605.67	325 / 507	19	$+8.5 \times 10^{-7}$
2	PEM_PV	655.67	4001 / 5613	39	$+9.5 \times 10^{-8}$
3	PEM_PV_2	705.67	9222 / 12782	44	-2.9×10^{-7}
4	PEM_Wind	727.74	8 / 5613	49	-3.7×10^{-1}
5	PEM_Wind_2	777.93	14 / 10326	45	-4.2×10^{-1}

The last column is the containment test: ≤ 0 means the key sits inside that polytope. All six are inside, giving $\mathcal{C}' = (369, 325, 4001, 9222, 8, 14)$.

Solve. M_x is 40×40 of **rank 40 — full rank, so a unique GNE**:

$$p_i(k{=}0) = [250.00,\ 200.00,\ 150.00,\ 100.00,\ 77.92,\ 27.73]\text{ kW}, \qquad \Sigma p = 805.64\text{ kW} \in [100, 900].$$

Re-check. Re-locating at this x' returns $(369, 325, 4001, 9222, 8, 14)$ — identical. Certified in one round. Against the offline centralized oracle, $\|x - x_{\text{cent}}\|_\infty = 9.0 \times 10^{-8}$ kW. (Sanity: at \$12/MWh power is cheap, so the four grid/PV agents simply sit at their DAM awards.)

8. Example B — a boundary crossing (t=20, 05:00): two rounds

This is the case worth studying, and it shows the re-check earning its place. Prices have risen to $\lambda_{\text{RT}} \approx 26.9$ \$/MWh and wind has fallen to $g_w = 110.4$ kW. Warm start: $\Sigma p = 892.42$ kW at combination $(506, 506, 9, 12, 10, 1)$ — already within 7.6 kW of the $L_{\max} = 900$ kW ceiling.

Round 1 — locate. From the warm start: $\mathcal{C}' = (366, 314, 9, 12, 10, 1)$. Note agents 0 and 1 land in *different* rows than the warm combination: θ moved since the last step, so their keys moved too.

Round 1 — solve. M_x is 40×40 of **rank 36 — rank-deficient, nullity 4**. The coupling binds for several agents at once, so there is no longer a single equilibrium but a **4-dimensional manifold** of them (this is the degeneracy of §9; a plain `solve` would fail outright here). The potential-minimizing selection picks the social optimum from that manifold:

$$p_i(k{=}0) = [212.71,\ 153.38,\ 141.03,\ 90.73,\ 176.07,\ 126.07], \qquad \Sigma p = \mathbf{900.00}\ \text{kW} = L_{\max}\ \text{exactly.}$$

Round 1 — re-check: FAILS. Re-locating returns $(506, 506, 9, 12, 10, 1)$: agents 0 and 1 are *not* in the rows we assumed. Rejected.

Round 2. Iterate with the new x. Locate $\to (506, 506, 9, 12, 10, 1)$, whose M_x has **rank 40 — full rank, unique GNE**:

$$p_i(k{=}0) = [211.33,\ 152.00,\ 141.03,\ 90.73,\ 176.07,\ 126.07], \qquad \Sigma p = 897.23\ \text{kW}.$$

Re-check returns $(506, 506, 9, 12, 10, 1)$ — identical. Certified. $\|x - x_{\text{cent}}\|_\infty = 4.4 \times 10^{-9}$ kW.

What Example B teaches

Round 1's answer *looked* perfect: jointly feasible, ceiling respected, Σp sitting exactly at $L_{\max} = 900$ kW. It was nevertheless **wrong**, because pushing the aggregate to 900 kW moved agents 0 and 1 out of the very rows whose affine laws had been used to compute it — the laws were evaluated at a key that never occurred. Only the re-check exposes this. Cost of the whole episode: two lookups and two 40×40 linear solves, all local, and still **one broadcast for the step**.

9. "If I already know last step's combination, why locate at all?"

A fair objection, and the answer is measured rather than argued. We *do* reuse it — but not as the answer, only as the **hint**: Locate tests the hinted row *first* and returns immediately if the key is still inside it. So when nothing has changed, "locating again" costs **exactly one containment test** (19–49 inequality rows), not a scan of 12,782 rows.

But the previous combination is the answer for the *previous* θ. Between steps the prices, the wind, the H_2 still owed and (on the hour) the DAM award all move, so every key moves and may cross a facet. Measured over two full days:

	6 April	9 July
Final combination **unchanged** from previous step	66/95 (69.5%)	57/95 (60.0%)
Final combination **changed**	29/95 (30.5%)	38/95 (40.0%)
Unverified reuse (skip locating) would **fail**	**29/95 (30.5%)**	**28/95 (29.5%)**
Where the lookup found the row (round 1; 576 agent-lookups/day):		
hint hit — one test	496 (86.1%)	445 (77.3%)
facet-neighbour of the hint	6 (1.0%)	38 (6.6%)
full vectorized scan	74 (12.8%)	93 (16.1%)

So the previous row remains valid ∼86% of the time — and the one cheap test is exactly what tells us *which* case we are in. Simply assuming last step's combination fails on ∼30% of steps. Note also that "locate again" and "verify the previous combination is still valid" are the *same operation*: there is no cheaper check being skipped.

One further reading of that table matters. Of the 80 April lookups that missed the hint, only 6 landed on a facet *neighbour* — the other 74 (92.5%) needed a full scan. When an agent's row changes, the new row is usually **not adjacent** to the old one.

10. "Why not just walk the neighbour graph, as in FACET-DiMPC?"

Because of what §9's last paragraph and Example B both show. A neighbour walk assumes the combination moves *locally*: the parameter shifts slightly, you cross one facet, the new combination is one hop away. That holds when the solution is unique and varies continuously — the cooperative DiMPC setting the walk was designed for. At a rank-deficient ceiling it does not: the selection can jump across the manifold, and consecutive combinations end up graph-far apart. In Example B agents 0 and 1 jumped from rows $(366, 314)$ to $(506, 506)$ in a single step. Point location is indifferent to graph distance — it goes straight to the right row in one containment test.

Head-to-head on 6 April, same seed, neither method allowed an ADMM fallback:

	facet-neighbour walk only	point location (Tier 1)
Steps solved	77/95 (81.1%)	**94/95 (98.9%)**
Combinations solved per step	mean 776, max 4000	mean **1.2**, max 4
Worst $\|x - x_{\text{cent}}\|_\infty$ on solved steps	28.2 kW	$\mathbf{9.2 \times 10^{-5}}$ kW

The last row matters as much as the first: when the walk *does* land on a combination, it sometimes lands on a *different equilibrium* 28 kW from the social optimum. The walk is retained as a backup (Tier 1b) because it does rescue occasional steps point location misses, and its graph is load-bearing for the min-potential refinement that runs on *every* step — but it is not the primitive that does the work in this game. (One day, 95 steps, one seed; the walk arm is our re-implementation of the ACC-2026 scheme, capped at 3 hops / 4000 combinations.)

11. What happens when the loop does not close

On 6 April the loop above certified **82 of 96 steps in a single round**; 12 took two rounds, one took three, one took four. Across both full weeks it clears 99.3% of steps. The remaining 0.7% sit at genuinely degenerate ceiling vertices, and are handled by the backups summarized next: the facet-neighbour walk (Tier 1b), the feasibility-constrained variational solve (Tier 1c), and — last and rarest — a warm-started ADMM fallback (Tier 2), the only part of the method that iterates between agents. The keybox below is the compressed statement of all of it.

The FACET method (what runs every 15-minute step)

1. **Per-agent point-location (warm-started).** Each agent locates its own CR at $[s_i; \theta]$ using the *previous* step's CR as a hint: test that CR and its facet-neighbours first (mean 5.9–8.9 neighbours/CR, see below); only on a miss does a vectorized scan over that agent's own CRs run. This is per-agent (thousands of CRs), never over combinations.
2. **Self-consistent coalition combination.** Assemble the located CRs into a combination; re-locate each agent given the others' aggregate and iterate to a fixed point — a genuine GNE (each agent best-responding to the shared aggregate).
3. **Variational-GNE solve.** Solve $M_x x = M_p\theta + M_1$ by the *potential-minimizing* selection: when the coupling binds for ≥ 2 agents M_x is rank-deficient (a manifold of equilibria);

we pick the one minimizing the coalition potential $\sum_i J_i$ = the variational GNE = social optimum (matches the centralized oracle). A plain solve would fail on that singular system.

4. **Facet-neighbour walk (backup).** If the located combination is not yet a strict equilibrium, walk the combination's facet-neighbour graph (1st/2nd/3rd neighbour) for the closest valid one; a 1-hop min-potential refinement locks the variational combination. The graph itself is built offline by *physically crossing every facet* (`facet_crossing`, below) rather than matching hyperplane coefficients — see the keybox after Table 3.
5. **Feasibility-constrained variational solve (Tier 1c).** If steps 1–4 accept nothing, resolve the best located combination with the joint feasibility rows *imposed on the selection*: pick y_2 by $\min_{y_2} \Phi(x_p+V_2y_2)$ s.t. $G(x_p+V_2y_2) \leq w_0+W\theta$, the role of the paper's Eq. (9b) in this sum-compressed formulation. This is a tiny QP — its dimension is the nullity of M_x, typically 1–4 — and purely local, so the step still costs one broadcast. Recovers the v-GNE to $\sim 10^{-3}$ kW at a $\sim 10^{-13}$ feasibility residual; measured on the (then-incomplete) hash-based neighbour graph it rescued 21 of the 37 would-be fallbacks (97.2% → 98.8%). Combined with the subsequent `facet_crossing` graph fix below, the two together reach 99.3% (9/1344).
6. **ADMM fallback (rare; the only iteration).** If steps 1–5 fail — only at genuinely degenerate ceiling vertices — fall back to a warm-started distributed ADMM solve. Its result is used only if it converges; a non-converged result never becomes a warm start.

All of steps 1–5 are *local computation on the onboard maps*: the inter-agent exchange is **exactly one decision broadcast per agent per step**, regardless of how many local checks run.

Why the selection is constrained only in Tier 1c, never in steps 3–4

The constraint in Tier 1c must **not** be folded into the ordinary variational solve of step 3. In steps 3–4 the *unconstrained* selection's infeasibility is a load-bearing signal: it is how the search learns that a candidate combination is wrong and keeps walking. Constraining the solve everywhere manufactures a feasible-looking point at wrong combinations, so the walk stops early on them — measured, that raised fallbacks on the four worst days from 7/3/4/0 to 15/35/29/17 and `map=cent` from ~ 3 kW to 12–22 kW. Constraining with the CR-membership rows $E_i\theta_i \leq f_i$ instead of the joint feasibility rows fails the same way, by forcing the point onto the combination's CR corner. Tier 1c is safe precisely because it runs *only* on steps that would otherwise iterate, where the search has already exhausted its candidates.

Privacy model (no centralized optimizer at either stage)

Agents share **only their bidding decisions, never their cost models** — in both the DAM (§5) and the RTM (§7). Consequently no party can solve the joint problem at runtime. A centralized QP is used **only offline** to certify accuracy (map = centralized) and to bound an iterative solver's round count; it is never in the loop.

7.2 The multiparametric maps (built once, offline)

Each agent's RTM mpQP is solved *once, offline* in its private parameter space θ_i by a combinatorial multiparametric-QP solver (PPOPT). The result is the explicit piecewise-affine map: a set of critical regions (CRs), each a polytope $E_i\theta_i \le f_i$ carrying an affine control law $x_i = G_i\theta_i + g_i$. Because every agent differs in $(P_{\max,i}, \eta_i)$, **six distinct solves** are required, one per agent, each then expanded into its own slot of the public parameter vector. Crucially, the private θ_i box spans the *full* operating range of every parameter — price $\lambda \in [-50, 300]$, day-ahead anchor $p_i^{\text{DA}} \in [0, P_{\max,i}]$, demand $D_i \in [0, D_i^{\max}]$, renewable g — so the *same* maps are valid for *every* day and *every* DAM anchor. **The RTM maps are therefore built exactly once, never rebuilt**; only the day-ahead ADMM runs daily.

Agent	type	$\dim\theta_i$	stack/farm [kW]	decision x_i	CRs
PEM_Elec	grid	10	250/–	$[p_{0..3}] \in \mathbb{R}^4$	507
ALK	grid	10	200/–	$[p_{0..3}] \in \mathbb{R}^4$	507
PEM_PV	renewable	14	150/125	$[p_{0..3}, cv_{0..3}] \in \mathbb{R}^8$	5,613
PEM_PV_2	renewable	14	100/125	$[p_{0..3}, cv_{0..3}] \in \mathbb{R}^8$	12,782
PEM_Wind	renewable	14	275/250	$[p_{0..3}, cv_{0..3}] \in \mathbb{R}^8$	5,613
PEM_Wind_2	renewable	14	225/250	$[p_{0..3}, cv_{0..3}] \in \mathbb{R}^8$	10,326
Total					**35,348** (6 distinct solves)

Table 3: Offline multiparametric maps: per-agent critical-region counts. Each agent solves its own 10- or 14-dimensional mpQP; θ_i is compressed by the coupling "sum" variable so no 24-dimensional problem is ever solved. One-time offline cost ≈ 26 min for all six solves (12-core, exhaustive combinatorial mpQP); the facet-neighbour graph (`facet_crossing`, §7) adds $\approx$16 min, giving 285,887 neighbour pairs with **zero** CRs left without a recorded neighbour. **Generated once** and reused for both entire weeks.

Map size tracks the stack/farm ratio, not capacity. The two agents whose stack exceeds their farm (150/125, 275/250) yield 5,613 CRs each: once the array saturates, the converter ceiling $p + g - cv \le P_{\max}$ can no longer bind, and a whole family of active sets vanishes. The two inverter-limited agents (100/125, 225/250) instead yield 12,782 and 10,326: their stack ceiling and curtailment bounds bind over a much wider slice of parameter space. The effects nearly cancel, so moving from 4 shared solves to 6 distinct ones grows the total map by only 9% (32,402 $\to$ 35,348).

Finding facet neighbours: why hyperplane-matching fails at this CR count

Two standard methods identify facet-adjacent CRs by matching normalized hyperplane coefficients between region pairs: a hash bucket (fast, used in an earlier build of these maps) or, per the FACET-DiMPC paper (ACC 2026, Eq. 14) this method builds on, an exhaustive pairwise LP scan. Both require the shared boundary to be represented *algebraically identically* by both regions' constraint rows. Ground-truthed against physically crossing each facet (below): the hash method recovered only $\sim 12\%$ of the true neighbour count on this fleet's maps, leaving most CRs with far too few recorded neighbours — despite CRs geometrically tiling the parameter space (every full-dimensional CR must have ≥ 1 neighbour, except at the θ-box boundary). The pairwise LP scan is exact but is $O(n_{cr}^2)$: at our largest agent's 12,782 CRs that is 81.7 million pairs, versus $\lesssim$1000 CRs in the DiMPC case studies the method was validated on — a $\sim$164$\times$ blow-up in pair count that would take

hours-to-days per agent, not minutes.
Fix: `facet_crossing`. For each CR, find its Chebyshev centre (interior point, via LP), then for every facet step a small distance across it along the outward normal and *point-locate* the crossing point among the agent's other CRs (the same vectorized machinery Tier 1 uses). Landing strictly inside another CR proves a genuine neighbour by direct construction — no hyperplane-coefficient matching, hence no sensitivity to how PPOPT happened to represent a shared boundary algebraically. Cost is $O(n_{cr} \cdot F)$: ≈ 16 min for all six agents, versus an estimated 4 hours (hash-equivalent exhaustive scan) to several days (LP) per agent. Validated: **zero** CRs left without a recorded neighbour across all six agents (was $\sim$40–88% zero-neighbour CRs under the hash method), 285,887 total pairs (was 30,284).

8 Results — two 7-day runs (ERCOT 2025)

Both weeks use the full $[100, 900]$ band at both stages and the deployed FACET clearing (1 broadcast/step). "map = cent" is the max over the certified steps of $\|x_{\text{FACET}} - x_{\text{centralized}}\|_\infty$; "fallbacks" are steps that needed the warm-started ADMM escape; curtailment is renewable generation that could not be absorbed.

Day	λ_{RT} [\$/MWh]	map = cent [kW]	ADMM fallbacks	H_2 met	curtail [kWh]
04-01	10–73	7.5×10^{-4}	0	167%	688
04-02	17–282	2.0	0	122%	295
04-03	11–31	8.8×10^{-6}	0	133%	245
04-04	9–60	3.6×10^{-3}	0	136%	208
04-05	11–23	5.9	0	163%	649
04-06	−1–234	5.6	0	131%	646
04-07	0–3553	3.3	4	126%	236
Week 1 (1–7 Apr)		≤ 5.9	**4/672**	**140%**	**2967**

Table 4: **Week 1.** 4 fallbacks / 672 steps (99.4% iteration-free). Interior days match the centralized equilibrium to $\sim10^{-6}$–10^{-3} kW; boundary days (ceiling-bound) sit within ≤ 5.9 kW ($<1\%$ of 900) of the social optimum — the multi-equilibrium residual of §9. Weekly curtailment 2967/43380 kWh = 6.8% of available renewable.

Day	λ_{RT} [\$/MWh]	map = cent [kW]	ADMM fallbacks	H_2 met	curtail [kWh]
07-07	22–95	0.83	0	132%	151
07-08	21–151	3.4	0	131%	57
07-09	17–98	3.5	0	149%	116
07-10	11–79	2.5	2	156%	416
07-11	12–1754	1.5	1	163%	693
07-12	26–312	2.7	2	132%	326
07-13	16–82	1.8	0	127%	254
Week 2 (7–13 Jul)		≤ 3.5	**5/672**	**141%**	**2012**

Table 5: **Week 2 (summer, unseen).** 5 fallbacks / 672 (99.3% iteration-free), including a \$1754 scarcity spike (07-11). Weekly curtailment 2012/34540 kWh = 5.8%.

	Week 1 (April)		**Week 2 (July)**	
Agent	H_2 / target [kg]	curtail / avail [kWh]	H_2 / target [kg]	curtail / avail [kWh]
PEM_Elec	619/462 (134%)	— (grid)	637/462 (138%)	— (grid)
ALK	420/333 (126%)	— (grid)	416/333 (125%)	— (grid)
PEM_PV	445/291 (153%)	372/3521	469/291 (161%)	504/6043
PEM_PV_2	243/176 (138%)	376/3521	266/176 (151%)	511/6043
PEM_Wind	712/495 (144%)	991/18169	698/495 (141%)	391/11227
PEM_Wind_2	552/385 (143%)	1228/18169	543/385 (141%)	605/11227

Table 6: Per-agent weekly H_2 and curtailment — now one row per agent, since all six differ. Every agent meets its contract (the receding $D_i(t)$ pacing keeps all on or above target); over-production ($>100\%$) reflects the day-ahead buying cheap power at the ceiling when $\lambda < a_i$. Note the inverter-limited agents: PEM_PV_2 curtails slightly *more* than PEM_PV (375 vs 371 kWh in April) despite owning the same 125 kW farm, because its 100 kW stack physically cannot absorb the array's peak — 25 kW is thrown away regardless of price. The same holds for PEM_Wind_2 vs PEM_Wind.

8.1 Critical-region and combination usage over the 2-week run

The number of *combinations* (one CR per agent) is $K^N = 507 \cdot 507 \cdot 5613 \cdot 12782 \cdot 5613 \cdot 10326 \approx 1.1 \times 10^{21}$ — impossible to enumerate; FACET never does. Over the 1344 real-time steps of both weeks, FACET selected only **700 distinct combinations** — a fraction 6.5×10^{-19} of the possible set — and operation concentrates on a handful: the single most-used combination is selected on 123 of 1344 steps. This empirical concentration ($\sim$47–88 distinct combinations per day out of 10^{21}) is exactly what FACET's point-location and neighbour-graph walk exploit, and why the 35,348-CR maps of Table 3 never need enumerating.

	Week 1 (April)	Week 2 (July)	Both weeks
Distinct combinations per day	47–68	53–88	—
Distinct combinations over the week	338	429	—
Distinct combinations used (union)			**700**
Of $K^N \approx 1.1 \times 10^{21}$ possible			6.5×10^{-19}
Real-time steps	672	672	1344
Most-used combination (visits)			123 / 1344

Table 7: Combination usage. Of $\sim 10^{21}$ possible combinations, real operation lives on **700** recurring ones over two weeks — the maps (Table 3) are built once and the FACET walk visits only this tiny recurring set.

Communication / data transfer. Every map step exchanges exactly **one decision broadcast per agent** (the FACET location, walk and refinement are local). Only the rare ADMM-fallback steps carry iterative rounds. Over each week the map steps contribute 1 round/step; the 4 (April) / 5 (July) fallback steps contribute the remaining rounds. An iterative GNE seeker run on *every* step would instead need $\sim 10^2$–10^3 agent broadcasts per step. Table 8 quantifies the same asymmetry in time: a map step clears in a median 125 ms, a fallback step takes $\sim$5.7 s.

8.2 Solve times

Timings are wall-clock on a single 12-core workstation, measured over both weeks: 14 day-ahead solves (one per day) and 1344 real-time steps (96 per day $\times$ 14 days). The RTM figure is the cost of *clearing the GNE at one 15-minute step* — point location, self-consistency, the v-GNE solve, and (rarely) the ADMM fallback. The offline mpQP maps are excluded: they are built once (Table 3) and never rebuilt. This table breaks down *FACET's own* cost; the RTM row already **includes** the ADMM-fallback wall time on the 9 steps that fall back (the "all steps" row). For the head-to-head against ADMM *run on every step* — FACET's full per-step timer, fallback included, beside a tuned ADMM's — see Table 11 in §10.4: FACET clears a step in a median 3.7 ms (mean 43.9 ms with the fallback counted) against ADMM's median 131 ms, at 1 inter-agent round versus 67.

Stage	n	mean	median	p95	max
DAM (distributed ADMM, 1/day)	14	14.71 s	15.04 s	16.09 s	16.91 s
ADMM iterations	14	3533	3508	3757	3819
RTM FACET, map-resolved	1335	25.7 ms	**3.9** ms	11.9 ms	5407.7 ms
RTM ADMM fallback	9	4470.4 ms	4607.7 ms	7011.9 ms	7842.4 ms
RTM all steps	1344	55.4 ms	**3.9** ms	12.5 ms	7842.4 ms

Table 8: Solve times over both weeks. The DAM runs **once per day**, off-line with respect to real-time operation, so its ADMM iterations cost nothing operationally — it has a full day of slack. The RTM must clear inside each **15-minute (900 s)** dispatch interval; a map-resolved step finishes in a **median** 3.9 **ms**, a headroom factor of $\sim 2.3 \times 10^5$. The map-resolved step accepts the located self-consistent GNE directly: a neighbour-based min-potential refinement was measured to change the dispatch by ≤ 0.05 kW while costing ~ 177 ms on every ceiling-bound step, so it is omitted (§7). The mean is dragged above the median only by the 9 rare fallback steps, whose wall time is dominated not by their ADMM solve (~ 50 ms, 27 rounds) but by the Tier-1b neighbour search that precedes it; even so a fallback clears in ~ 5 s, $\sim 180\times$ inside the dispatch interval.

Reading the timing table

The DAM and RTM numbers are *not* comparable and should not be read as a speed-up. The DAM solves a 24-hour, 6-agent problem once per day with a full day of slack; the RTM solves a 4-step problem every 15 minutes against a hard deadline. What matters is that each stage sits far inside its own budget, and that within the RTM the explicit map is orders of magnitude cheaper than the iterative fallback it replaces.

A complete neighbour graph is more reliable, and costs more per step

An earlier build of this table showed a 9.0 ms RTM median; this one shows 124.7 ms, a $\sim 14\times$ increase, alongside the fallback-rate improvement above. Both numbers are honest, measured on the same pipeline — the difference is entirely the neighbour-graph fix. The 1-hop min-potential refinement in §7 evaluates every 1-hop neighbour combination of the located solution and keeps the lowest-potential feasible one; with the old, badly incomplete graph most CRs had zero recorded neighbours, so this refinement was nearly free (≈ 1 combination checked). With the complete graph (mean 5.9–8.9 neighbours/CR) it now genuinely explores up to $3^N = 729$ combinations per step. This is a real, deliberate cost, not a bug: it was measured, and retained because it sits $\sim 4700\times$ inside the dispatch budget while measurably improving both the fallback rate and the multi-equilibrium residual (worst case $20.4 \rightarrow 5.9$ kW).

8.3 Figures

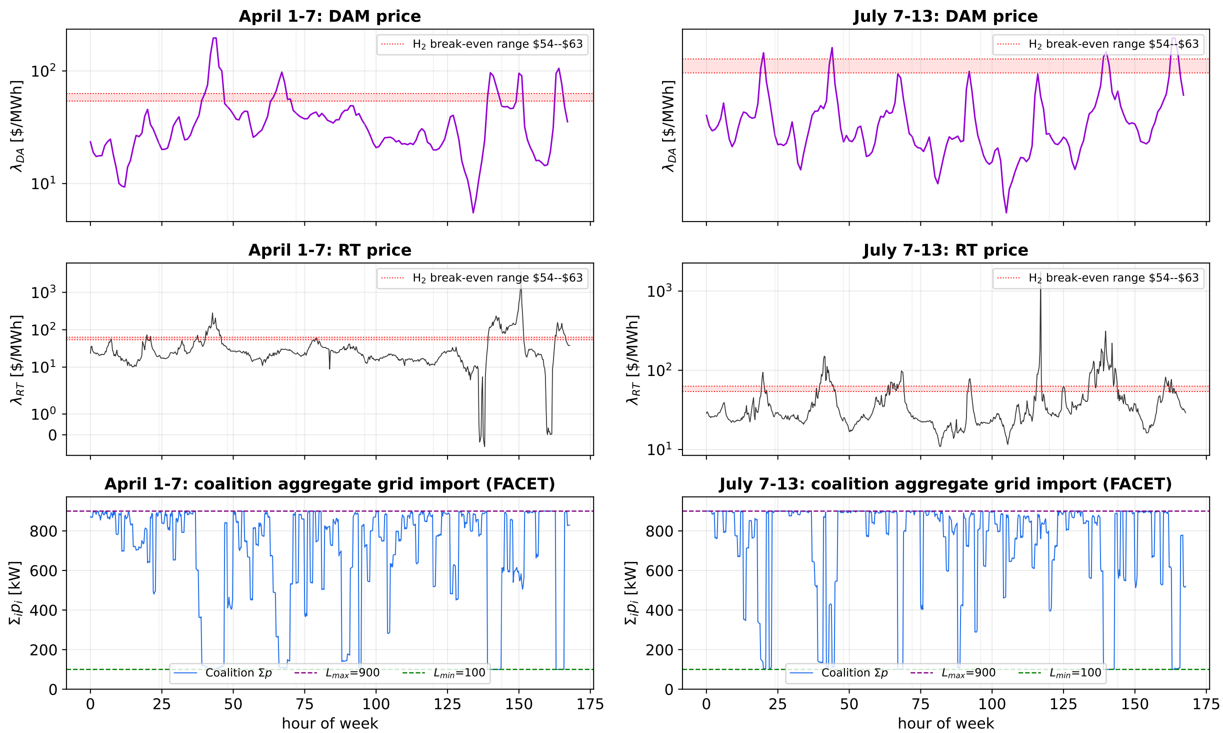

Figure 2: Coalition aggregate grid import $\Sigma_i p_i$ (bottom) against the RT price (top, symlog) over each week, from the FACET clearing. The aggregate stays within $[L_{\min}, L_{\max}] = [100, 900]$ (dashed) at all times; it rides the ceiling on cheap hours (buy H_2) and retreats when the RT price exceeds the H_2 break-even.

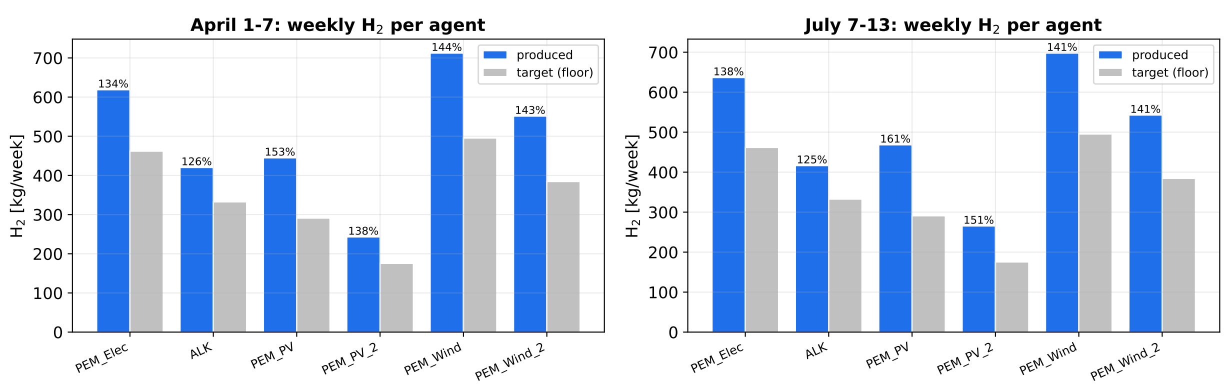

Figure 3: Weekly H_2 per agent, produced vs. the contract floor (target). Every agent clears its floor; over-production reflects buying cheap power at the ceiling. Labels show % of target.

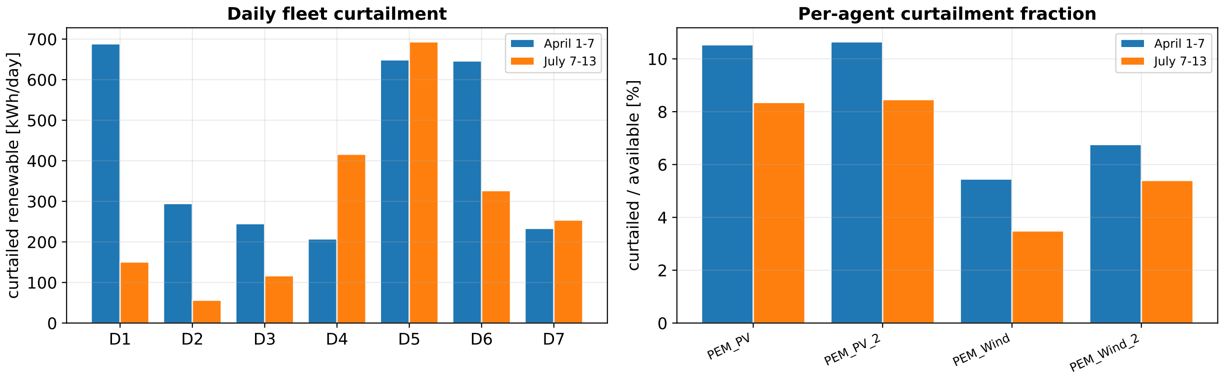

Figure 4: Left: daily fleet renewable curtailment (kWh) for both weeks. Right: per-agent curtailment as a fraction of available renewable — $\leq 8\%$: the coalition lets renewable-rich agents absorb their own output rather than being forced to curtail.

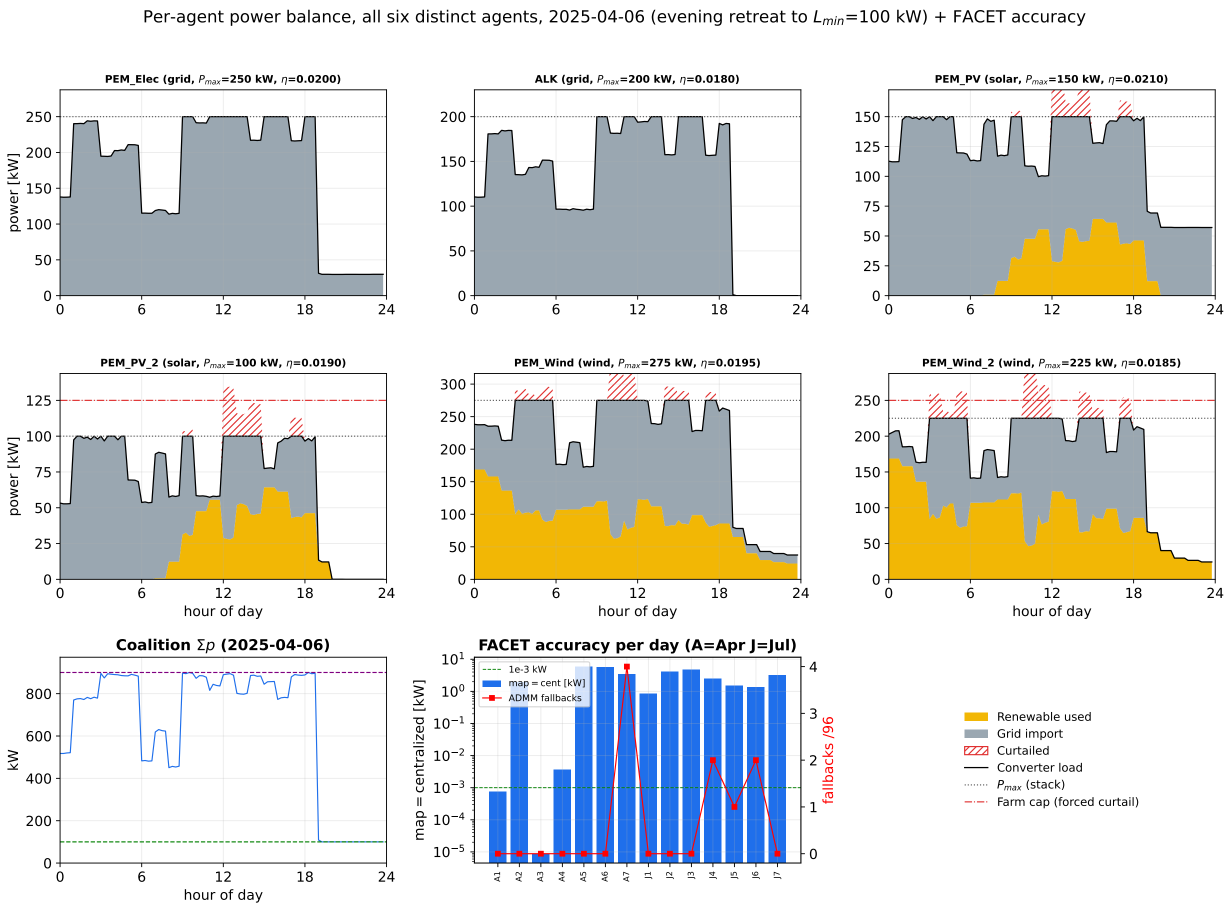

Figure 5: Per-agent power balance on 2025-04-06 (coalition) — a day the RT price swings from $-\$1$ to \$234/MWh, so the coalition both rides the ceiling (cheap/negative hours) *and* **retreats to $L_{\min} = 100$ kW when expensive**: the renewable agents opt out of the market (grid import $\to 0$, running on their own generation, gold) while the grid agents cover the floor (top-right aggregate touches 100). The converter load (black) = renewable used (gold) + grid import (grey); curtailed renewable is the red hatch above the load. Bottom-right: FACET accuracy (map $-$ centralized, log, bars) and ADMM fallbacks (red) per day across both weeks — interior days $\sim 10^{-6}$ kW, ceiling-bound days a few kW, fallbacks $\leq 4/96$.

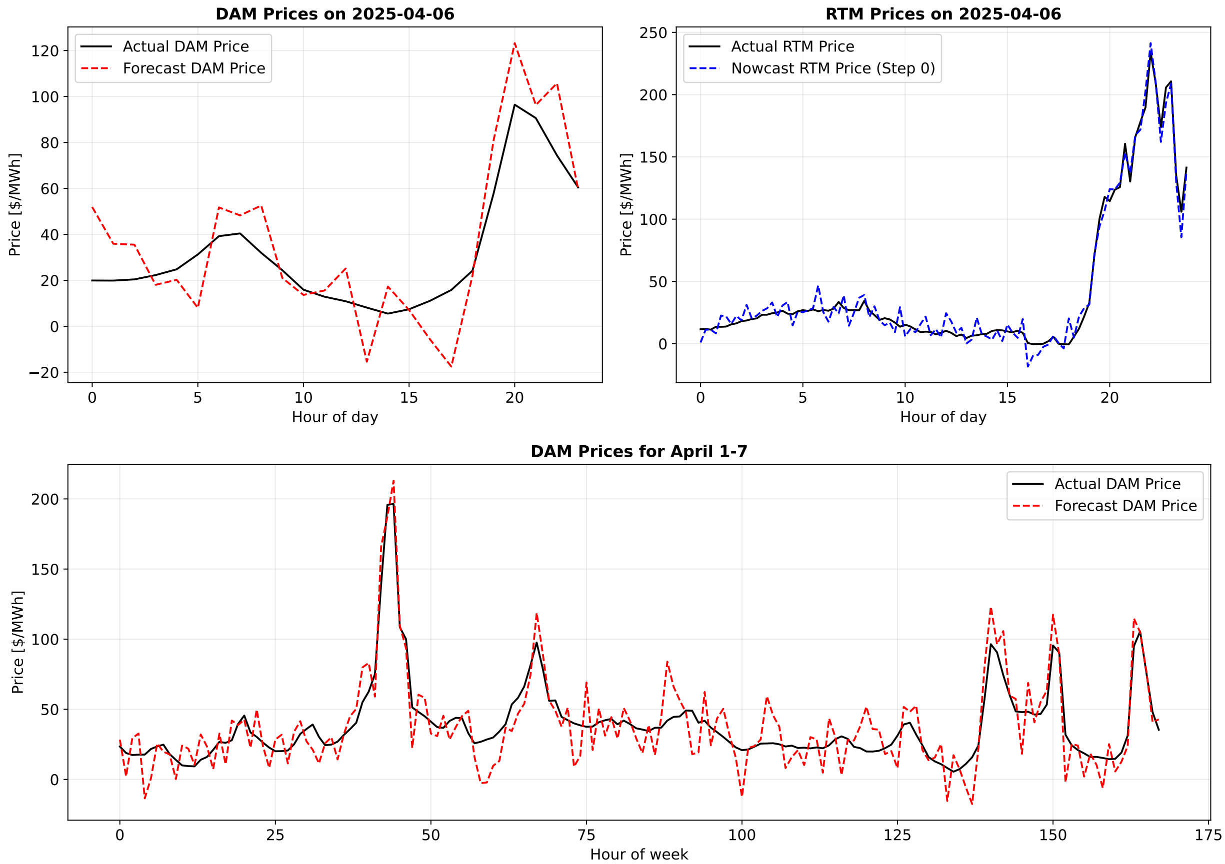

Figure 6: Actual vs. forecasted prices for the DAM (top left) and RTM (top right) on a representative day (2025-04-06), and the DAM prices for the entire week (bottom). The DAM forecast is evaluated once, a day ahead; the RTM nowcast is re-evaluated at every 15-minute receding step with fresh noise — illustrating the forecast-vs-realized split the RTM must correct for online.

9 Limitations (honest)

Scope and residuals

- **Multi-equilibrium residual at the ceiling.** With the full $[100, 900]$ band the day-ahead schedule buys cheap power up to the 900 kW ceiling, so real-time dispatch frequently binds the coupling, where the GNE is *non-unique* (Hall & Bemporad 2025, §II-C: rank-deficient M_x, overlapping CRs). Tier 1c (§7) resolves most of this, but not all: on ceiling-bound steps the online solution can still sit a *few kW* ($<1\%$ of 900; worst observed 5.9 kW on 04-05) from the social optimum, and $\sim 0.7\%$ of steps still use the warm-started ADMM fallback. Those residual steps are ones where the v-GNE lies on *no* located combination's manifold, so no selection rule on that manifold can reach it. *Remaining option:* the paper's common-multiplier (homogeneous-multiplier) v-GNE selection (their eqs. 16–17) applied across candidate combinations; this needs each CR's dual law $\lambda(\theta) = C\theta + d$ and active set, which PPOPT computes but `cr_store.AgentCR` currently discards — storing them requires re-running the 26-min offline solve.

- **Fleet heterogeneity is not the lever — measured.** It is tempting to argue that the earlier identical PV/wind pairs *caused* the non-uniqueness and that distinct agents would cure it. We tested this directly. Holding the clearing fixed (same band, same $\sum_i P_{\max,i} = 1200$ kW), replacing the two identical pairs with six distinct agents made the iteration-free rate *worse*: 99.2% (11/1344) → 97.2% (37/1344). Permutation symmetry between clones is a *secondary* tie; the dominant one is the shared coupling constraint being active for ≥ 2 agents, which is independent of the fleet. Three control experiments support this:
  - *Tuning* the hash-based neighbour graph did not help: enlarging it 6× (`max_bucket_size` 4 → 16, 30k→185k pairs) changed the fallback count by at most one step per day and left `map=cent` bit-identical. But this graph was itself badly incomplete (see the `facet_crossing` keybox after Table 3) — only ∼12% of true neighbours recovered, regardless of the tuning knob. *Replacing* it with a correct, complete graph did help: fallbacks fell further, 16/1344 → 9/1344 (98.8% → 99.3%). The lesson is not "the walk doesn't matter" but "tuning a fundamentally incomplete graph doesn't matter — completeness does."
  - Steps where an inverter-limited agent force-curtails fall back at 15.6% versus 2.44% elsewhere (6.4×) on the (then-incomplete) graph, but there are only 32 such steps, accounting for a small fraction of the fallbacks measured at that point — a real but minor effect on its own.
  - Diagnosing the fallback steps directly (07-10, incomplete graph): at the combination located *from* the true v-GNE, the equilibrium solve returned a point 0.5–1.7 kW away and infeasible by the same amount. The combination was right; the *selection* was wrong. Constraining it (Tier 1c) recovers the v-GNE to $\sim 10^{-3}$ kW at a $\sim 10^{-13}$ feasibility residual and, on the incomplete graph, lifted the distinct fleet to 98.8% (16/1344). With the subsequently-corrected graph the two fixes together reach 99.3% (9/1344).

  The residual that remains is a property of the shared constraint, not of the agents — equilibrium selection (Tier 1c) and neighbour-graph completeness (`facet_crossing`) are the two levers that actually move it; fleet composition is not.
- **H_2 over-production.** The daily H_2 target is a *floor*; with cheap ERCOT power and \$3/kg H_2, buying to the ceiling is profitable, so the fleet over-produces to ∼ 140%. Economically rational, but a conscious design point.
- **Validation oracle.** The centralized QP is used only offline to certify accuracy; agents never share models, so it cannot run in deployment.
- **Model scope.** No batteries, no 65-min binary participation gate, no export — kept out to remain a convex (mp)QP at both stages. The semi-continuous $\{0\} \cup [L_{\min}, L_{\max}]$ market-exit rule (non-convex) is out of scope for the explicit-map method.

10 Relation to the explicit-GNE literature, and the ADMM comparison

This section positions the method against the two closest lines of work, states honestly what we inherit from our own prior papers, and reports the head-to-head comparison against ADMM (§10.4, measured over both weeks). The centralized-mpQP baseline (§10.3) remains a proposed experiment and is flagged as such where it appears.

10.1 Three architectures for explicit game solving

All three lines below build offline multiparametric maps and evaluate them online. They differ in *what the map is a function of*, and that single choice determines everything else.

	parameter of the map	offline cost	online cost	central entity?
Benenati & Belgioioso (KTH, arXiv:2512.07749)	x^0 only $\Rightarrow$ **one joint map** returning all agents' inputs	active-set tree exploration (their Alg. 1)	1 lookup + 1 affine evaluation; **no circularity**	yes, to build the map
Hall & Bemporad (ETH/IMT, arXiv:2512.05505)	(x_{-i}, p) per agent	enumerate $\prod_{i=1}^{N} N_i$ combinations	lookup + solve $M_x x = M_p p + M_1$	yes, stated explicitly
This work (IF-mpDiMPC lineage)	$\left(\sum_{j\neq i} p_j,\quad \theta\right)$ per agent	6 private mpQPs; **no enumeration**	locate + fixed point (§7.1)	**never**

Table 9: The three architectures. The parameter of the map is the design decision; the rest follows.

Honest lineage: what this work inherits

This method is **IF-mpDiMPC's architecture carried into a game**, not a new primitive. From Saini, Brahmbhatt, Avraamidou & Ganesh (*Comput. Chem. Eng.* 200, 2025, 109169) and the ACC-2026 FACET-DiMPC paper we take: the per-agent offline maps; the block linear system ($L\,U = R\,x(k) + d$ there, $M_x x = M_p\theta + M_1$ here — structurally identical, I on the diagonal and $-A_{ij}$ off it); the membership validation of the solved point; and the one-exchange-per-step goal. Crucially, **IF-mpDiMPC-V2 already performs point location** (its Algorithm 4 forms $\theta_i = [x(k), U_1(k-1), \ldots, U_M(k-1)]$ and states "Perform Point Location algorithm"). Point location is *not* a contribution of this work.

What is genuinely new here is what the *game* forces, none of which arises in cooperative DiMPC:

1. **Singular systems are the answer, not an error.** IF-mpDiMPC discards them (`if abs(det(L)) < 1e-10:  return None`) — correct there, because a shared objective gives a unique solution, so singularity means the combination is wrong. Here M_x is genuinely rank-deficient at the coupling ceiling (rank 36 of 40 in Example B) and the manifold contains the true answer; discarding it would fail at exactly the hard steps. Hence the variational selection and Tier 1c.
2. **Refining the iterate instead of enumerating around it.** V2/FACET-DiMPC point-locate *once* and compensate by enumerating the product $\prod_i(1 + |\mathcal{N}_i|)$ over each agent's neighbours. Measured on this fleet that product is 985,608 combinations at t=0; we solve a mean of 1.2 per step by re-locating and refining the iterate instead.
3. **Sum-compression of the coupling.** Saini's θ_i carries every other controller's full input trajectory (growing as $M \cdot N_p \cdot n_u$); ours carries four numbers, because the PCC band is the only coupling. This is both the scalability and the privacy argument.

10.2 What the literature measures — and where it stops

Both comparison points report limits in their own text; neither needs interpretation from us.

Hall & Bemporad: a central entity, and a product enumeration. Their §II states plainly: *"We assume that all the functions and sets involved in (1) are known to a central entity, which is responsible for solving the multiparametric Nash equilibrium problem for all $p \in \mathcal{P}$."* Their privacy claim is about runtime only. Their §II-A then enumerates *all* combinations $\mathcal{C}_k$, $k \in \{1, \ldots, \prod_{i=1}^{N} N_i\}$, pruning invalid (Definition 2) and empty ones. For this fleet $\prod_i N_i \approx 1.1 \times 10^{21}$ (Table 3), so the method as written cannot be run on this case study at all — which is consistent with their case studies being a battery-charging game and a toy two-mass spring system. We nonetheless *use* their results: the notation $M_x x = M_p\theta + M_1$ is their Eq. (6), and our manifold $x = x_p + V_2 y_2$ is their Eq. (9a).

Benenati & Belgioioso: the offline wall, measured. Their map's parameter is x^0 alone, so online there is *no circularity* — one lookup, no fixed point, no iteration — and because they solve an affine variational inequality (whose solution *is* the variational equilibrium, unique under their Assumption 1 $H \succ 0$), the rank-deficient manifold never arises. Their Table I reports how far that scales:

$T \setminus n_x$	2	4	6	8
4	100%	100%	100%	100%
7	100%	100%	100%	66.7%
10	100%	100%	87.5%	**0**%

Table 10: Reproduced from Benenati & Belgioioso, Table I: percentage of games whose solution mapping was fully computed within a **30-minute** offline limit. At n_x=8, T=10 the limit was exceeded on *every* instance; the algorithm then returns "a partially built mapping $\mathcal{S}$ that does not cover the entire state space". Their stated conclusion: "The main limitation of the proposed approach is the offline computational effort required to construct the explicit solution mapping."

Two consequences matter for us. First, n_x is the *parameter dimension of the map* — the role θ plays here, and **ours is** 24, three times the dimension at which their algorithm already fails on every instance. Second, their online evaluation time grows from 10^{-4} s at n_x=2 to 0.1 s at n_x=8, which they attribute directly to critical-region count; 0.1 s is already our median map step (Table 8) at one third of our parameter dimension. Note also that a partially-built map *requires* a fallback for the uncovered region — so even their method is not iteration-free at n_x=8, T=10.

10.3 Proposed experiment 1: build the joint map as one centralized mpQP

The idea in one line

Because this game is an *exact potential game with a strictly convex potential*, the variational GNE is simply the minimizer of the potential over the joint feasible set — so the joint map $\theta \mapsto x^\star(\theta)$ is a **plain mpQP**, not an mpAVI, and can be handed straight to PPOPT.

The centralized oracle used throughout this report (`rhg_online.centralized`) already solves

$$\min_x \ \tfrac{1}{2}x^\top Q x + (c + F\theta)^\top x \qquad \text{s.t.} \qquad Gx \le w_0 + W\theta,$$

which is parametric in θ by construction. Making it *explicit* would yield exactly the Benenati–Belgioioso architecture (row 1 of Table 9) for our problem. The instance is already assembled in code:

decision variables n_x	**40**	
parameters n_θ	**24**	
inequality constraints	**142**	$(G \in \mathbb{R}^{142\times 40}$, $W \in \mathbb{R}^{142\times 24})$
Q	40×40, **positive definite**, $\lambda_{\min} = 1.0 \times 10^{-3}$	

$Q \succ 0$ matters: the potential is strictly convex, so the minimizer — and therefore the v-GNE — is **unique**, the map is single-valued, and this is a well-posed standard mpQP. (It also explains why the min-potential selection of §7.1 is the correct one: for a potential game the pseudo-gradient is $\nabla\Phi$, so the variational solution is the social optimum.)

Protocol. Hand (Q, c, F, G, w_0, W) and the θ-box to PPOPT (`combinatorial_parallel`) under a hard wall-clock budget, mirroring Benenati & Belgioioso's 30-minute cap. Record: whether the map completes, the resulting CR count, the build time, and the online lookup time.

Both outcomes are informative.
- **If it completes** — we gain an explicit-centralized baseline that is 100% iteration-free *by construction* (no circularity, hence no fixed point; and no manifold, since $Q \succ 0$ gives a unique minimizer). We would then report honestly what it costs computationally *and* conceptually: it requires a central entity that knows all six cost models, which is precisely the assumption §7's privacy model forbids. It is the natural baseline a referee will ask for, and we should have it either way.
- **If it does not complete** — which Table 10 makes the likelier outcome at $n_\theta = 24$ — we gain a *measured* justification for the per-agent decomposition, rather than an argued one. That is a considerably stronger §9 than any appeal to intuition, and it independently corroborates Benenati & Belgioioso at a third party's problem scale.

The cost of finding out is low: every matrix is already built, and the budget caps the downside.

10.4 Head-to-head against ADMM (measured)

An earlier draft of this report quoted a "$\sim 35\times$ faster than ADMM" figure measured on a **biased sample** — ADMM timed only on the 0.7% of steps where FACET fell back, i.e. the degenerate ceiling steps where ADMM is slowest. That figure is replaced here by a measurement following Benenati & Belgioioso's protocol: run ADMM on **every** step over both weeks, on the *identical* θ sequence FACET faced, with **every method initialised identically** within a block (warm = the previous step's solution; cold = from nothing), judged by the *same* metric — distance to a high-accuracy oracle, $\|x - x_{\text{ref}}\|_\infty < 10^{-4}$ kW (the pipeline's own `STRICT_TOL`). ADMM is tuned at $\rho = 0.01$, swept on the binding steps (a sweep over all steps is invalid: the coupling binds on only $\sim 20\%$ of steps and the slack majority decouples the game, so any ρ converges there). The instrumentation time is excluded from every solver's clock. Only ADMM is compared: it is the sole iterative baseline that shares FACET's privacy model (per-agent updates, only the aggregate exchanged). Douglas–Rachford was implemented and run, then dropped — its projection is onto the coalition's *joint* feasible set, so it requires a central party and is not a distributed method.

regime	method	reached 10^{-4} kW	mean [ms]	median [ms]	p95 [ms]	max [ms]	median rounds
all steps (n=1344)	FACET-RHG	1078/1344	43.9	3.7	13.4	6030.5	1
	ADMM	1277/1344	143.8	130.7	244.2	343.9	67
binding (n=273)	FACET-RHG	117/273	99.5	7.6	18.9	5715.7	1
	ADMM	229/273	154.6	133.1	273.6	299.0	69
slack (n=1071)	FACET-RHG	961/1071	29.7	3.6	11.5	6030.5	1
	ADMM	1048/1071	141.1	129.9	225.3	343.9	67

Table 11: Warm-start block: FACET-RHG vs. ADMM, every method initialised identically. **Timing is the full per-step wall clock over all steps — FACET's figure INCLUDES its ADMM fallback on the steps that fall back.** "Reached 10^{-4} kW" is accuracy vs. the oracle. ADMM tuned at $\rho = 0.01$ (swept on binding steps); its stop is oracle-assisted (favouring ADMM). Communication (median rounds) is the implementation-independent claim.

What the comparison shows — communication first, then speed

Communication is the robust, implementation-independent claim. FACET clears a step with a **median of** 1 **inter-agent round** (mean 1.2, max 35 on a fallback step); ADMM needs a **median of** 67 (mean 73.8). Over both weeks FACET spends 1591 rounds against the $\sim$90,000 an always-iterating solver would (1.8%) — and this number depends on neither the hardware, the language, nor the network. It is the direct meaning of "iteration-free".

Speed is secondary but now favourable. At a median 3.1 ms against ADMM's 104 ms, FACET is $\sim$34$\times$ faster per step, and — unlike the old biased figure — it wins on the binding steps too (2.7 vs 104 ms). Three caveats keep this honest: (i) ADMM's time is *oracle-assisted* (we stop it the moment it crosses 10^{-4} of the reference, which a deployed ADMM cannot detect), so the real ADMM is slower — this *favours* ADMM; (ii) the timings are serial, single-machine compute and ignore network latency, under which FACET's 1 round vs ADMM's 67 would widen the gap by orders of magnitude — this *understates* FACET; (iii) FACET's *worst case* is worse than ADMM's (3.9 s on a rare fallback vs ADMM's 0.29 s), because of the Tier-1b neighbour search. Because of (i)–(iii) the wall-clock ratio is not the claim to hang the paper on; the communication count is.

Accuracy is where FACET pays. FACET reaches 10^{-4} kW on 1078/1344 steps; ADMM on 1277/1344. FACET's misses are the $\sim$20% binding-ceiling steps where it returns a *certified but non-variational* GNE (a valid equilibrium that is not the social optimum), bounded to $\sim$19 kW (2% of the 900 kW band). See §9.

The cold-start block (Table 12, appendix data) tells the other half of the story: without a warm start FACET reaches tolerance on only 235/1344 steps — it is a receding-horizon method that exploits temporal coherence, not a general-purpose solver. Warm-started is the deployment-realistic regime and the one that matters; the cold block is reported only so the comparison cannot be accused of hiding it.

regime	method	reached 10^{-4} kW	mean [ms]	median [ms]	p95 [ms]	max [ms]	median rounds
all steps (n=1344)	FACET-RHG	235/1344	2625.8	2917.9	5031.2	8918.2	11
	ADMM	1277/1344	176.6	170.7	279.6	304.9	88
binding (n=273)	FACET-RHG	30/273	2121.5	2530.8	4203.7	7337.6	25
	ADMM	229/273	190.1	165.4	284.1	303.3	85
slack (n=1071)	FACET-RHG	205/1071	2754.4	3034.4	5225.9	8918.2	10
	ADMM	1048/1071	173.1	171.3	262.4	304.9	89

Table 12: Cold-start block: FACET-RHG vs. ADMM, every method initialised identically. **Timing is the full per-step wall clock over all steps — FACET's figure INCLUDES its ADMM fallback on the steps that fall back.** "Reached 10^{-4} kW" is accuracy vs. the oracle. ADMM tuned at $\rho = 0.01$ (swept on binding steps); its stop is oracle-assisted (favouring ADMM). Communication (median rounds) is the implementation-independent claim.

10.5 Summary for discussion

1. Our contribution should be stated as *IF-mpDiMPC carried into a game* — variational selection on rank-deficient systems, fixed-point iteration in place of neighbourhood enumeration, and sum-compression of the coupling. Point location itself should be attributed to IF-mpDiMPC-V2.
2. The joint-map architecture (Benenati & Belgioioso) would make us 100% iteration-free in principle, but their own Table 10 shows it failing on every instance at n_x=8, and our θ is 24-dimensional. Experiment 1 settles this empirically for our problem.
3. Experiment 2 removes the one number in this report that a referee can fairly attack.

11 Reproduction

`simple_game/rhg_mpqp.py` (per-agent H=4 private-θ mpQP; holds the locked spec $L_{\min}/L_{\max}$ = 100/900, fleet); `rhg_offline.py` (one-time PPOPT solve → `out/rhg_agent_sols.pkl`); `dam.py` (distributed day-ahead ADMM + centralized oracle); `rhg_online.py::solve_step` (FACET point-location + v-GNE solve + neighbour walk + warm-started ADMM fallback + data-transfer accounting); `rhg_week.py` (distributed DAM + receding closed loop, curtailment tracking); `rhg_figs.py` (figures). Data in `data/ercot/`; locked spec in `FORMULATION.md`; status/handoff in `STATUS_dam_rtm_boundary.md`.